

Problem 1. What is the definition of an *orthogonal basis* of an inner product space V ?

A set $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ is called an orthogonal basis for V if it is a basis (i.e., if it is an independent, spanning set) and if for all $i \neq j$ we have $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$.

Problem 2. Consider the plane $V \subseteq \mathbb{R}^3$ defined by

$$V = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}.$$

I claim that V is a subspace and that $\{(1, 0, -1), (1, 1, 1)\}$ is an orthogonal basis for V .

(a) Compute the projection of the point $(1, 0, 0)$ onto V .

We note that these vectors are orthogonal with respect to the standard dot product:

$$(1, 0, -1) \bullet (1, 1, 1) = 1 + 0 - 1 = 0.$$

If $\mathbf{v}_1, \dots, \mathbf{v}_k$ is an orthogonal basis for a subspace $V \subseteq \mathbb{R}^n$ with respect to the standard dot product, then the orthogonal projection of a general point $\mathbf{x}$ onto V is

$$P_V(\mathbf{x}) = \frac{\mathbf{x} \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1} \mathbf{v}_1 + \dots + \frac{\mathbf{x} \bullet \mathbf{v}_k}{\mathbf{v}_k \bullet \mathbf{v}_k} \mathbf{v}_k.$$

In our case we let $\mathbf{x} = (1, 0, 0)$, $\mathbf{v}_1 = (1, 0, -1)$ and $\mathbf{v}_2 = (1, 1, 1)$, so that

$$P_V(\mathbf{x}) = \frac{\mathbf{x} \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{x} \bullet \mathbf{v}_2}{\mathbf{v}_2 \bullet \mathbf{v}_2} \mathbf{v}_2 = \frac{1}{2}(1, 0, -1) + \frac{1}{3}(1, 1, 1) = \frac{1}{6}(5, 2, -1).$$

(b) Compute the projection of the point $(1, 0, 0)$ onto the orthogonal complement $V^\perp$.

In general, we have $\mathbf{x} = P_V(\mathbf{x}) + P_{V^\perp}(\mathbf{x})$, hence

$$P_{V^\perp}(\mathbf{x}) = \mathbf{x} - P_V(\mathbf{x}) = (1, 0, 0) - \frac{1}{6}(5, 2, -1) = \frac{1}{6}(1, -2, 1).$$

This makes sense geometrically because the orthogonal complement $V^\perp$ is the line spanned by the normal vector $(1, -2, 1)$ of the plane $x - 2y + z = 0$, so the projection $P_{V^\perp}(\mathbf{x})$ must be a scalar multiple of $(1, -2, 1)$.

Alternatively, we could observe that $\mathbf{v} = (1, -2, 1)$ is a basis for $V^\perp$ (which is trivially an orthogonal basis), and then compute

$$P_{V^\perp}(\mathbf{x}) = \frac{\mathbf{x} \bullet \mathbf{v}}{\mathbf{v} \bullet \mathbf{v}} \mathbf{v} = \frac{1}{6}(1, -2, 1).$$

Here is a picture:

